

LiNbO₃ Material Properties on the ξ Power Ladder:

Phase Matching in PPLN:
Testing a Geometric Assignment

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March 2026

Contents

.1	Introduction: PPLN as the Core of the Coherent Ising Machine	3
	The DOPO Mechanism	3
	The Phase-Matching Condition	3
.2	The Exponent $\xi^{1/2}$ in T0 Theory	4
	$\xi^{1/2}$ as the Fundamental Electroweak Exponent	4
	Physical Meaning of the Exponent	4
	Distinction from Document 166	4
.3	Test: Δn versus $\sqrt{3} \cdot \xi^{1/2}$	5
	Numerical Comparison	5
.4	Further Matches on the ξ Ladder	6
	Ratio of Electro-Optic Coefficients	6
.5	$\xi^{1/2}$ as the Electroweak Rung	7
.6	Consequences for CIM Design	8
.7	Summary	9

Abstract

Periodically poled lithium niobate (PPLN) is the physical core of the Coherent Ising Machine (CIM): the $\chi^{(2)}$ nonlinear crystal implements, via parametric amplification, the DOPO spin — the fundamental bit of the Ising computer. The poling period Λ , refractive index $n(\omega)$, and electro-optic coefficient r_{33} are conventionally regarded as empirically determined material parameters with no geometric origin.

This document tests whether a central parameter of LiNbO_3 , the dispersion difference in DOPO operation, lies on the ξ -power ladder of T0 theory, namely in the form:

$$\Delta n_{(1064 \rightarrow 2128 \text{ nm})} \stackrel{?}{=} \sqrt{3} \cdot \xi^{1/2} = 0.020 \quad (1)$$

with $\xi = 4/30\,000$. This assignment does not hold: the Sellmeier equation after Zelmon 1997 (congruent LiNbO_3 , extraordinary) gives $n_e(1064 \text{ nm}) = 2.1555$ and $n_e(2128 \text{ nm}) = 2.1219$, i.e. $\Delta n = 0.0336$. Only this value matches the poling period: $\lambda_p/\Delta n = 1.064 \mu\text{m}/0.0336 = 31.6 \mu\text{m}$ ($\Lambda = 31 \mu\text{m}$), whereas 0.020 would require $\Lambda = 53 \mu\text{m}$. $\sqrt{3} \xi^{1/2} = 0.020$ therefore lies 41 % below the measured value. The exact equality $\sqrt{3} \cdot (4/30000)^{1/2} = \sqrt{12/30000} = 0.02$ is an algebraic identity; the measured value is known to only 2–3 digits.

In T0 theory the exponent 1/2 appears as the weak coupling constant $\alpha_W = \xi^{1/2}$, as the lepton-mass ratio $m_e/m_\mu \propto \xi^{1/2}$ and in further contexts (Docs. 041, 056, 061, 122, 133). The relations $E_{EW} \sim E_P \cdot \xi^{1/2}$ and $m_h/M_P = \xi^{1/2}$ are not tenable. The PPLN dispersion does not lie on this rung; what remains as a match on the ξ ladder is the ratio of the electro-optic coefficients $r_{33}/d_{33} \approx 1 + \xi^{1/4}$ (deviation 2.2 %).

.1 Introduction: PPLN as the Core of the Coherent Ising Machine

The DOPO Mechanism

The Coherent Ising Machine (CIM) solves the Ising Hamiltonian $H = -\sum_{ij} J_{ij} \sigma_i \sigma_j$ via a network of DOPO pulses (Degenerate Optical Parametric Oscillators) inside a fibre cavity.

The physical core is the $\chi^{(2)}$ nonlinear crystal: a pump photon at frequency ω_p is converted into two signal and idler photons at $\omega_p/2$. Above threshold the DOPO develops exactly two stable states — phase 0 or π — corresponding to the Ising spin $\sigma_i = \pm 1$.

The material of choice is periodically poled lithium niobate (PPLN). The question of why LiNbO_3 is usually answered empirically: largest known d_{33} coefficient combined with low absorption losses. This document tests whether a geometric answer follows from the ξ -power ladder.

The Phase-Matching Condition

In the quasi-phase-matching (QPM) scheme the phase mismatch $\Delta k = k_p - k_s - k_i \neq 0$ is compensated by periodically inverting the crystal domains with period Λ :

$$\Lambda = \frac{2\pi}{\Delta k} = \frac{\lambda_{\text{pump}}}{\Delta n} \quad (2)$$

The poling period is directly proportional to the inverse dispersion difference $\Delta n = n(\lambda_{\text{pump}}) - n(2\lambda_{\text{pump}})$ (λ_{pump} : pump wavelength for the DOPO, SH wavelength for SHG). For the DOPO, $\Delta k = 2\pi(n_p - n_s)/\lambda_p$, hence $\Lambda = \lambda_p/(n_p - n_s)$.

Process	Pump	Λ (μm)	Δn
OPO — DOPO operation (CIM)	1064 nm	31.0	0.034
SHG — Telecommunications	780 nm	19.5	0.040
SHG — Green laser	532 nm	6.8	0.078

Table 1: Typical PPLN parameters for LiNbO_3 (extraordinary polarisation, room temperature, Sellmeier coefficients after Zelmon 1997).

In detail, Sellmeier (Zelmon 1997, n_e) gives: DOPO $1064 \rightarrow 2128$ nm $\Delta n = 0.0336$, $\Lambda = 31.6$ μm ; SHG $1560 \rightarrow 780$ nm $\Delta n = 0.0405$, $\Lambda = 19.3$ μm ; SHG $1064 \rightarrow 532$ nm $\Delta n = 0.078$, $\Lambda = 0.532/0.078 = 6.8$ μm .

.2 The Exponent $\xi^{1/2}$ in T0 Theory

$\xi^{1/2}$ as the Fundamental Electroweak Exponent

In T0 theory, $\xi^{1/2} = (4/30\,000)^{1/2} = 0.011547 \dots$ is not merely one among many ξ powers but the fundamental transition exponent between the Planck scale and particle physics. It appears in a remarkable variety of independent contexts:

Physical quantity	T0 expression	Value	Doc.
Weak coupling constant	$\alpha_W = \xi^{1/2}$	1.15×10^{-2}	041, 061
Lepton-mass ratio	$m_e/m_\mu \propto \xi^{1/2}$	4.84×10^{-3}	056, 122
Bottom-quark mass	$m_b = v \cdot \frac{3}{2} \cdot \xi^{1/2}$	4.25 GeV	041
Planck constant (scaling)	$\hbar \sim \sqrt{\xi}$	—	105

Table 2: The exponent $\xi^{1/2} = 0.011547$ as the universal electroweak exponent in T0 theory. It marks the geometric transition from the Planck world to the particle-physics scale.

Two further $\xi^{1/2}$ relations do not hold: $m_h/M_P = \xi^{1/2}$ gives 1.15×10^{-2} , whereas the measured value is $m_h/M_P = 1.0 \times 10^{-17}$ (see Doc. 068); $E_{EW} \sim E_P \cdot \xi^{1/2}$ gives about 1.4×10^{17} GeV with $E_P = 1.22 \times 10^{19}$ GeV, not ~ 100 GeV, and is therefore not tenable as a derivation of the electroweak scale.

Physical Meaning of the Exponent

The exponent $1/2$ marks the *electroweak transition* on the ξ -power ladder:

$$\begin{array}{lll}
 \text{Planck scale:} & \xi^0 = 1 & E \sim E_P \\
 \text{Electroweak transition:} & \xi^{1/2} \approx 10^{-2} & \alpha_W, m_e/m_\mu \\
 \text{Particle scale:} & \xi^1 \approx 10^{-4} & m_e, \alpha_{EM}
 \end{array} \tag{3}$$

The PPLN dispersion does not lie on this rung (Section .3).

Distinction from Document 166

Document 166 uses the exponent $1/4$ for the Casimir scale ($L_\xi \propto \xi^{1/4}$, decomposition $41/4 = 10 + 1/4$), not $\xi^{1/2}$; there is no common exponent of the Casimir scale and the PPLN dispersion.

.3 Test: Δn versus $\sqrt{3} \cdot \xi^{1/2}$

Numerical Comparison

The dispersion difference of LiNbO_3 for the DOPO operation is obtained from the Sellmeier refractive indices for the extraordinary polarisation (Zelmon 1997):

$$\Delta n = n_e(1064 \text{ nm}) - n_e(2128 \text{ nm}) = 2.1555 - 2.1219 = 0.0336 \quad (4)$$

The expression of Eq. (1) gives:

$$\sqrt{3} \cdot \xi^{1/2} = \sqrt{12/30000} = 0.020000 \dots \quad (5)$$

	Value
Measured: $\Delta n(1064 \rightarrow 2128 \text{ nm})$	0.0336
T0 expression: $\sqrt{3} \cdot \xi^{1/2}$	0.0200

The assignment $\Delta n = \sqrt{3} \cdot \xi^{1/2}$ does not hold, because the expression lies 41 % below the measured value. The exact equality with 0.020 is an algebraic identity; the measured value is known to only 2–3 digits.

.4 Further Matches on the ξ Ladder

Ratio of Electro-Optic Coefficients

LiNbO_3 has two relevant coupling constants: the linear electro-optic (Pockels) coefficient $r_{33} = 30.8 \text{ pm/V}$ and the nonlinear coefficient $d_{33} = 27.2 \text{ pm/V}$. Their ratio:

$$\frac{r_{33}}{d_{33}} = \frac{30.8}{27.2} = 1.1324 \approx 1 + \xi^{1/4} = 1.1075 \quad (\text{dev. } +2.2\%) \quad (6)$$

$\xi^{1/4}$ is the sub-Planck geometry exponent in T0 (Doc. 009). The ratio of two electro-optic coupling constants lies on the ξ ladder — indicating that the crystal band structure itself is ξ -rational.

.5 $\xi^{1/2}$ as the Electroweak Rung

Domain	Quantity	T0 expression	Doc.
<i>Particle physics (Standard Model)</i>			
	Weak coupling	$\alpha_W = \xi^{1/2}$	041, 061
	Lepton masses	$m_e/m_\mu \propto \xi^{1/2}$	056, 122, 133
	Bottom quark	$m_b = v \cdot \frac{3}{2} \cdot \xi^{1/2}$	041
<i>Material properties (this document)</i>			
	EO coefficients	$r_{33}/d_{33} \approx 1 + \xi^{1/4}$	167
<i>Cosmology and quantum optics</i>			
	Casimir scale	$L_\xi = (15\xi/\pi^2)^{1/4} \hbar c / (k_B T_{\text{CMB}})$	078, 166
	H_0 ratio	$\hbar H_0 / m_e c^2 = (\pi/2) \cdot \xi^{10}$	165

Table 3: $\xi^{1/2}$ as the electroweak exponent and further ξ powers of T0 theory. The PPLN dispersion does not lie on the $\xi^{1/2}$ rung (Section .3).

The PPLN dispersion does not belong to this hierarchy: $\sqrt{3} \cdot \xi^{1/2}$ lies 41 % below the measured $\Delta n = 0.0336$. Likewise, the relations $E_{\text{EW}} \sim E_P \cdot \xi^{1/2}$ and $m_h/M_P = \xi^{1/2}$ are absent, because they are not tenable (Section .2).

.6 Consequences for CIM Design

A geometric answer to "Why LiNbO₃?" via $\Delta n = \sqrt{3} \cdot \xi^{1/2}$ does not hold, because the measured dispersion is 0.0336. The choice of material remains empirically founded: largest known d_{33} coefficient combined with low absorption losses. A selection rule $\Delta n = k \cdot \xi^{n/4}$ for CIM substrates based on this assignment, and predictions derived from it ($\sqrt{2} \cdot \xi^{1/2}$ at ~ 900 nm, $2 \cdot \xi^{1/2}$ at ~ 680 nm), therefore do not hold.

.7 Summary

1. **Result:** The dispersion of LiNbO_3 for the most important PPLN process ($1064 \rightarrow 2128 \text{ nm}$) is $\Delta n = 0.0336$ according to Sellmeier (Zelmon 1997). The assignment $\Delta n = \sqrt{3} \cdot \xi^{1/2} = 0.020$ does not hold: the expression lies 41 % below the measured value, and the equality with 0.020 is an identity.
2. **Exponent:** $\xi^{1/2}$ appears as $\alpha_W = \xi^{1/2}$, $m_e/m_\mu \propto \xi^{1/2}$ and in further quantities (Docs. 041, 056, 061, 122, 133). $E_{EW} \sim E_P \cdot \xi^{1/2}$ and $m_h/M_P = \xi^{1/2}$ are not tenable. With the Casimir scale, which carries the exponent 1/4 in Doc. 166, $\xi^{1/2}$ shares no exponent.
3. **Further match:** $r_{33}/d_{33} \approx 1 + \xi^{1/4}$ (dev. 2.2 %).
4. **Classification and conclusion:** The PPLN dispersion does not lie on the $\xi^{1/2}$ rung. The choice of material for CIM design remains empirically founded.

Related documents: Doc. 041 (Parameter derivation, $\alpha_W = \xi^{1/2}$), Doc. 056 (Universal derivation, lepton masses), Doc. 061 (CMB units, α_W), Doc. 081 (Summary, electroweak scale), Doc. 122 (Ratio vs. absolute, m_e/m_μ), Doc. 133 (Fractal correction), Doc. 161 (CIM fundamentals), Doc. 165 (H_0 ratio), Doc. 166 (Casimir-CMB- H_0 bridge).